

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients** in an Agile/Scrum team, processing **10,000+ monthly triage submissions** via Mayden iaptus REST/JSON integrations; contributed to **\$340K+** in revenue.
- Owned backend security and reliability for the eTriage stack: implemented **AWS KMS** encryption for clinical PII, hardened authentication and rate limiting, added integration tests, and remediated **20+ VAPT findings**.
- Participated in the research and resolution of customer-reported problems, monitoring production logs during releases and client integrations and shipping targeted fixes for tenant-specific incidents to keep the multi-tenant service stable.

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer on a provider-facing **SMART on FHIR** clinical decision support app (Next.js + Firebase/GCP) that surfaces covered, guideline-backed medication alternatives in the EHR prescribing flow to prevent prior authorization.
- Extending a **Firebase Cloud Functions** backend that builds alternatives from **RxNorm** normalization, formulary/prior-auth logic, and PubMed/guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility and coverage.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's CCR text-analysis platform (Python, **FastAPI**, sentence-transformers, React), productionizing a published NLP method into a self-serve web tool for researchers.
- Built a YAML model registry and a versioned construct library (**99 constructs from 38 questionnaires**) with a structured data-quality warning system; grew the backend suite to **40 hermetic tests** running in CI without ML dependencies.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ automated tests** and provider fallback.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built an edge-native AI research assistant end to end on Cloudflare Workers (TypeScript): stateful WebSocket chat via Durable Objects and durable multi-step workflows via WorkflowEntrypoint with D1 storage, so tasks survive process restarts and individual tool failures.

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Built a cross-platform desktop app (React, Electron, FastAPI) with a fully offline on-device pipeline (MediaPipe, whisper.cpp, local TTS), designing async APIs and queues to keep the UI responsive while handling local video and audio streams.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Java, Python, TypeScript, JavaScript, C++, SQL, Bash, MongoDB, PostgreSQL, SQLite

Backend & APIs: Node.js, Express, FastAPI, React, REST APIs, JSON, microservices, async services

Testing & Agile: pytest, unit / integration testing, LambdaTest, secure coding, CI/CD, GitHub Actions, Agile / Scrum

Cloud & DevOps: Docker, Kubernetes, Linux, Git, AWS, GCP, Azure

ML & AI: PyTorch, sentence-transformers, Hugging Face Transformers, LLM agents, RAG, evaluation

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform with **Stripe** payments and a partial-payment ledger; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.